

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – [College Admission Agent (RAG Based)]

Domain of Project – [Education]

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Problem Statement -

Problem Statement - College Admission Agent (RAG Based)

The Challenge - A College Admission Agent, powered by RAG (Retrieval-Augmented Generation), streamlines the student admission process.

It retrieves and summarizes admission policies, eligibility criteria, and FAQs from institutional databases and official sources. Prospective students can ask natural language questions and receive accurate, up-to-date responses instantly.

The agent helps with course selection, application guidance, fee structure, and important deadlines.

Using trusted, real-time data, it reduces manual inquiries and enhances applicant experience.

This AI-driven assistant boosts transparency, accessibility, and efficiency in college admissions.

Technology - Use of IBM cloud lite services /IBM Granite is mandatory.

Technology Used

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models

I used LangChain +

Proposed Solution -

The proposed solution is an AI-powered College Admission Agent designed to assist prospective students by providing instant responses to admission-related queries. The system is developed using IBM watsonx Assistant, where admission FAQs, eligibility criteria, fee details, required documents, scholarship information, and important deadlines are configured as intents, dialog flows, and knowledge sources.

The chatbot is deployed through IBM Watson Assistant Web Chat and integrated into a College Admission website using the generated embed script. Students can interact with the virtual assistant through a user-friendly web interface and receive real-time responses to common admission questions.

The solution operates 24/7, reducing the workload on admission staff and eliminating delays caused by office hours. By providing consistent, accurate, and easily accessible information, the system improves the admission experience for students and parents.

The architecture is designed to be scalable and can be extended in the future with Retrieval-Augmented Generation (RAG), ChromaDB, IBM Granite models on watsonx.ai, and WhatsApp integration through Twilio for broader accessibility.

Langflow component Used -

Chat Input – Receives admission-related queries from students through the college admission website chat interface.

File Component – Stores admission FAQs, eligibility criteria, fee structure, required documents, and scholarship information used by the AI assistant.

HuggingFace Embeddings – Converts admission documents into vector embeddings to enable semantic search and efficient information retrieval.

Chroma Vector Store – Stores vectorized admission data and retrieves the most relevant information based on student queries.

Retriever Component – Searches the knowledge base and fetches relevant admission information to answer student questions accurately.

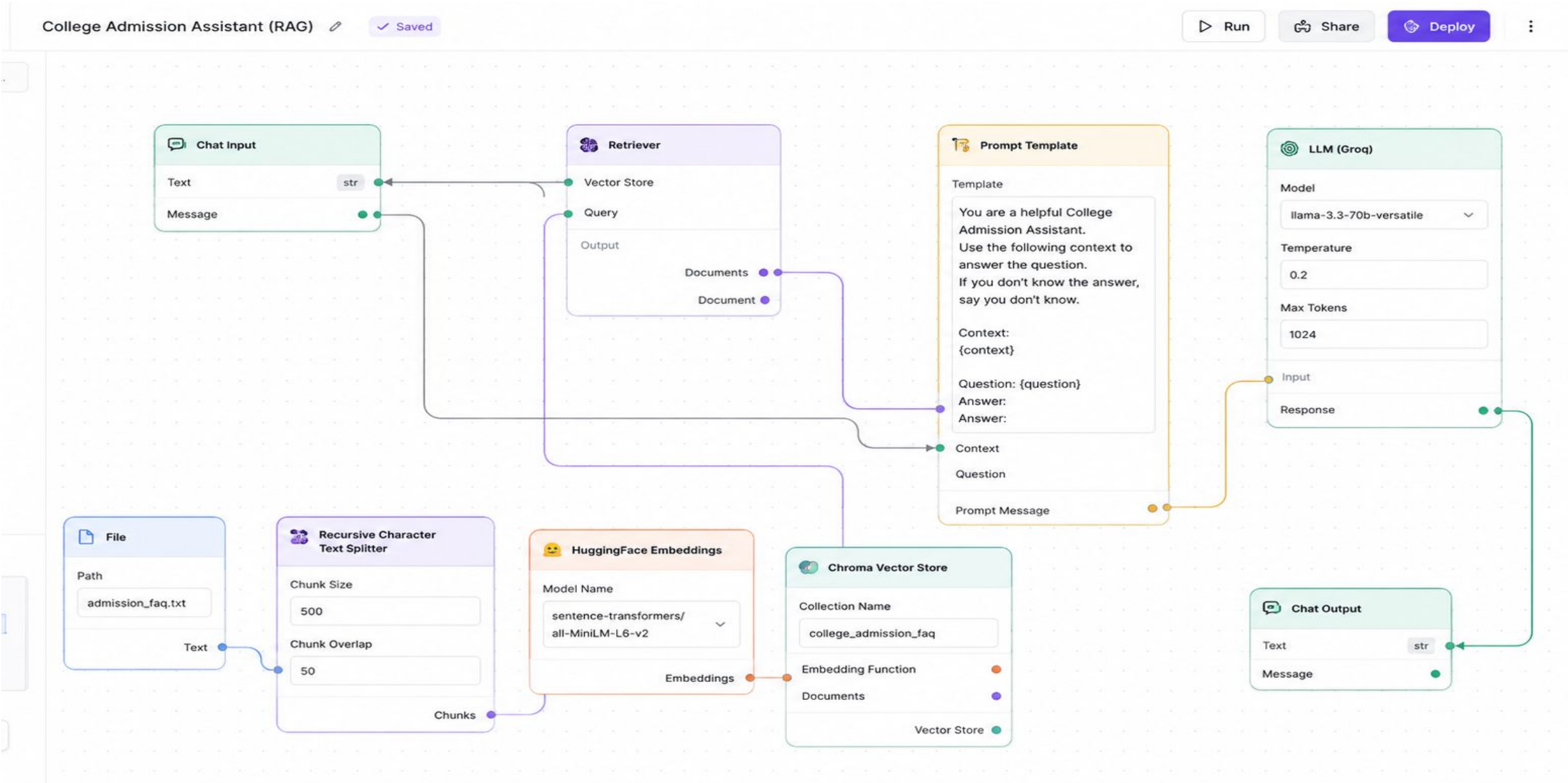
Prompt Template Component – Combines the student's query with retrieved context before sending it to the language model.

IBM Granite Model – Generates accurate, context-aware responses for admission-related questions using IBM watsonx.ai.

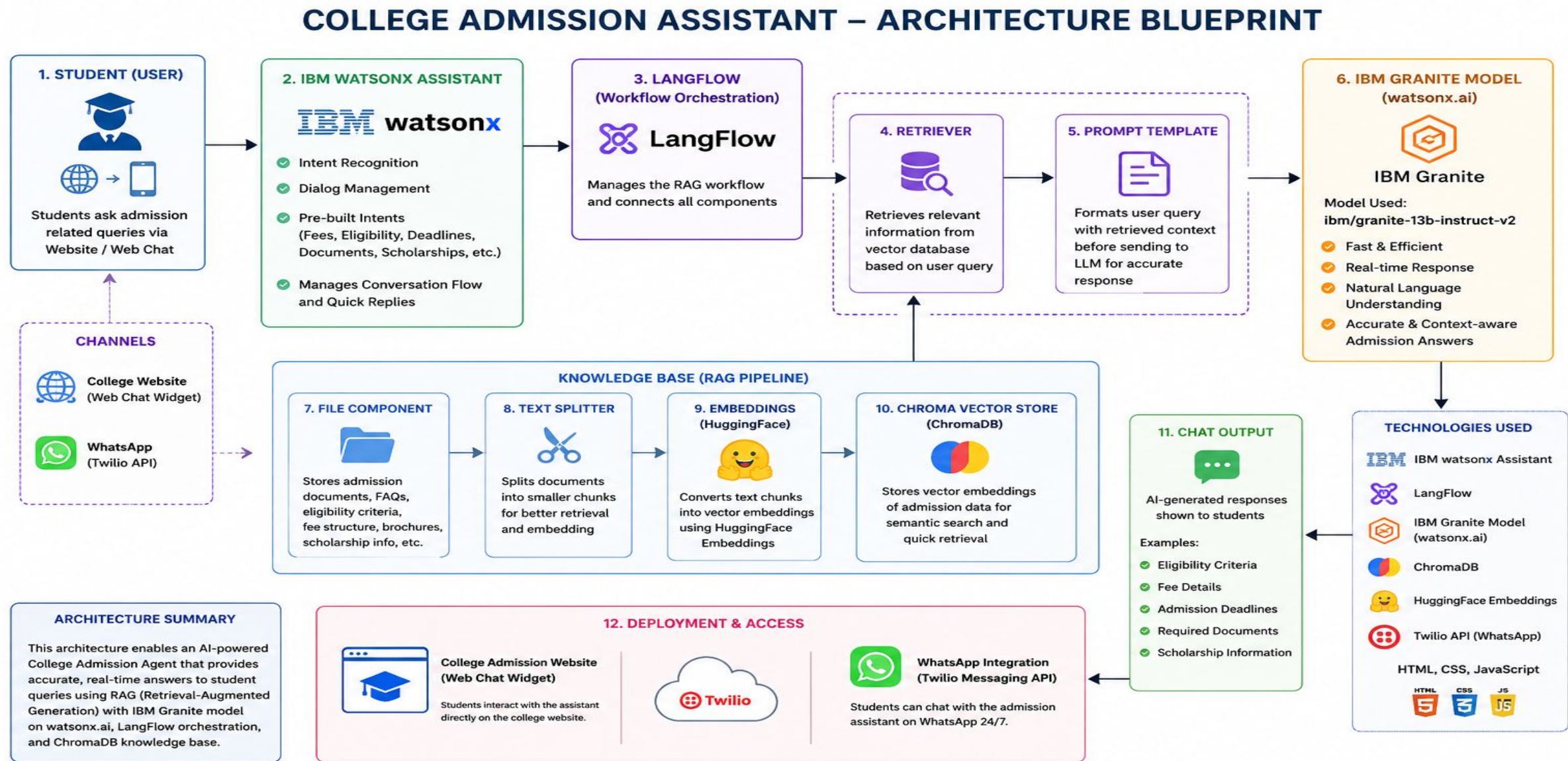
IBM watsonx Assistant – Provides the conversational interface and manages interactions between students and the admission agent.

Chat Output – Displays responses such as eligibility criteria, fee details, admission deadlines, and document requirements.

Langflow workflow -

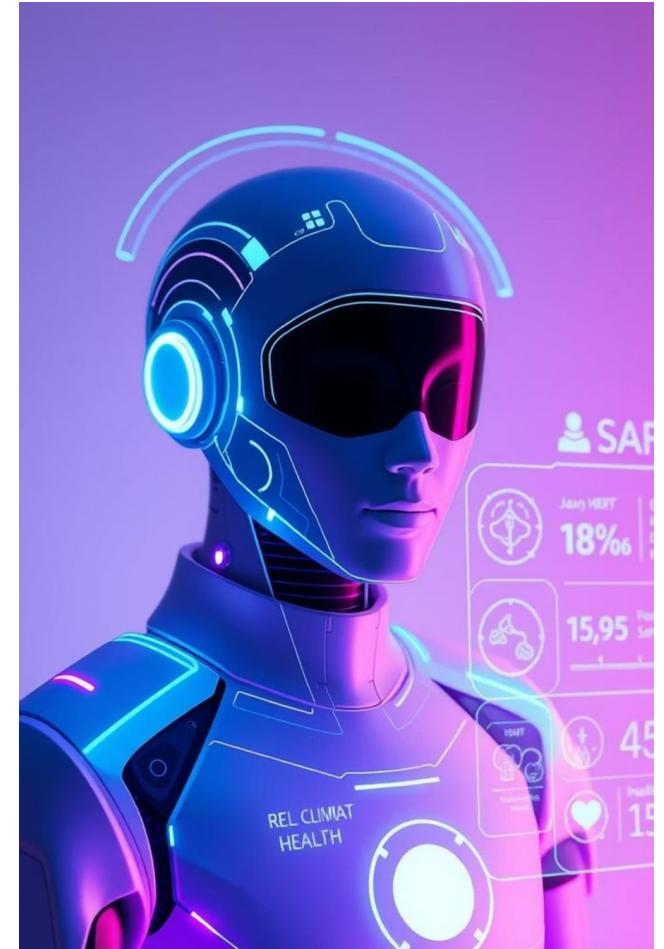


Architecture blueprint



Role of Agentic AI in the solution

Agentic AI enables the College Admission Assistant to autonomously understand student queries, retrieve relevant admission information from the knowledge base, and generate accurate responses without human intervention. It can manage multi-turn conversations, guide students through the admission process, provide personalized assistance based on user queries, and ensure 24/7 availability of admission support. By combining IBM watsonx Assistant, LangFlow, RAG, and IBM Granite Models, the agent can reason over available information, make context-aware decisions, and deliver reliable admission guidance efficiently.



Technology Used

- **IBM watsonx Assistant** – Conversational AI interface for handling student interactions and admission queries.
- **IBM Granite Model (watsonx.ai)** – Large Language Model used for generating accurate and context-aware responses.
- **LangFlow** – Visual workflow builder used to design and orchestrate the RAG pipeline.
- **ChromaDB** – Vector database used to store and retrieve admission-related information.
- **Hugging Face Embeddings** – Converts admission documents into vector embeddings for semantic search.
- **Retrieval-Augmented Generation (RAG)** – Enhances response accuracy by combining retrieved knowledge with AI-generated answers.
- **HTML, CSS, JavaScript** – Used to develop the College Admission Assistant website.
- **IBM Watson Assistant Web Chat** – Embedded chatbot interface integrated into the website.

Novelty and Uniqueness

1. **AI-Powered Admission Guidance** – Provides instant, intelligent, and personalized responses to student admission queries 24/7.
2. **RAG-Based Knowledge Retrieval** – Uses ChromaDB and LangFlow to retrieve accurate information from admission documents before generating responses.
3. **IBM Granite Integration** – Leverages IBM Granite Models on watsonx.ai for context-aware and reliable admission assistance.
4. **Agentic AI Approach** – Acts as a virtual admission counselor that understands queries, retrieves relevant information, and guides students through the admission process autonomously.
5. **Seamless Website Integration** – Integrated with IBM watsonx Assistant Web Chat, allowing students to access admission support directly from the college website without requiring human intervention.

Git Hub Link

- <https://github.com/Developer-123/College-Admission-Agent>

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Future Scope

1 WhatsApp Integration – Enable students to access the admission assistant directly through WhatsApp using Twilio.

2. Advanced RAG Knowledge Base – Expand the knowledge base with brochures, admission policies, previous FAQs, and academic regulations for more accurate responses.

3.Student Application Tracking – Allow applicants to check admission status, document verification status, and important updates in real time.

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Thank You!

Thank you for your time and interest.